

AP Precalculus Functional Skills Add-on Subdivision (20 problems)

1. Divide and write as $Q(x) + \frac{R(x)}{D(x)}$: $\frac{5x^2 - 3x + 7}{x - 2}$.

Show your work here

2. Divide: $\frac{2x^3 + 7x^2 - 5x + 6}{x + 1}$.

Show your work here

3. Divide: $\frac{6x^3 - 4x^2 + 9x - 1}{3x - 2}$.

Show your work here

4. Use division to rewrite $\frac{x^3 - 1}{x - 1}$. Then state the end behavior of this rational function.

Show your work here

5. Write $\frac{x^3 + 2x^2 - 1}{x}$ as a polynomial plus a proper fraction; state the end behavior.

Show your work here

6. Find the x -intercepts of $f(x) = \frac{(x-1)(x-4)}{x^2-9}$ (after simplifying if needed).

Show your work here

7. Find all real zeros of $g(x) = \frac{(x+5)(x-2)^2}{(x-2)(x+1)}$ and state whether each intercept is a “cross” or “bounce.”

Show your work here

8. Determine the x -intercepts of $h(x) = \frac{(x^2-4x+3)}{(x-3)}$ after simplifying; classify as cross/bounce if relevant.

Show your work here

9. True/False (justify): If a factor cancels, the corresponding x -value cannot be an x -intercept of the simplified rational function.

Show your work here

10. State all vertical asymptotes of $r(x) = \frac{x^2 - 1}{(x - 1)(x - 4)}$ after any simplification.

Show your work here

11. Find the vertical asymptotes of $q(x) = \frac{(x - 3)^2}{(x - 3)(x + 2)}$.

Show your work here

12. Does $s(x) = \frac{x^2 - 16}{x + 4}$ have any vertical asymptotes? Explain using factorization.

Show your work here

13. Find the vertical asymptotes of $t(x) = \frac{(x + 1)(x - 2)}{x(x - 2)^2}$ (simplify first if helpful).

Show your work here

14. Locate any hole(s) for $F(x) = \frac{(x-6)(x+2)}{(x+2)(x-1)}$ and give the simplified function.

Show your work here

15. Identify the hole(s) (if any) and simplified form for $G(x) = \frac{x^2 - 9}{x^2 - 6x + 9}$.

Show your work here

16. Determine if $H(x) = \frac{(x-4)^2}{(x-4)}$ has a hole; if so, where? Is there a vertical asymptote at that x ?

Show your work here

17. State the end behavior (HA or slant) of $P(x) = \frac{4x^2 - 7}{2x^2 + 5}$.

Show your work here

18. Determine the end behavior of $Q(x) = \frac{3x^3 - x + 1}{x^2 - 1}$ (name the slant if it exists).

Show your work here

19. Determine the end behavior of $R(x) = \frac{2x - 5}{x^3 + 1}$.

Show your work here

20. Use long division (or reasoning) to find the end behavior of $S(x) = \frac{x^4 - 3x^2 + 1}{x^2 - 2}$.

Show your work here